



Bangladesh University of Engineering and Technology

Course No.: ME 310

Course Title: Thermo-Fluid System Design

Project Title: Gasketed-Plate Heat Exchanger Design.

Group: **B23**

Section: B2

Submitted to:

Priom Das

Lecturer, Department of Mechanical Engineering, BUET.

Submitted by:

1. 1910091-Sabit Hasnat Mahee
2. 1910114-Md. Shahriar Parvez
3. 1910115-Barkat Ullah Fahad
4. 1910117-Arman Mia

TABLE OF CONTENTS

Acknowledgement.....	1
Abstract.....	2
Chapter 1: Introduction.....	3
1.1 Introduction to Gasketed-Plate Heat Exchanger	3
1.2 Problem Statement.....	4
1.3 Objectives.....	4
Chapter 2: Overview of the Project	5
2.1 Specifications	5
2.2 Operating Principles of a Gasketed-Plate Heat Exchangers	5
2.3 Advantages of Gasketed-plate Heat Exchanger.....	5
2.4 Limitations and Possible Solutions	7
Chapter 3: Calculation & Optimization.....	9
3.1 Assumptions.....	9
3.2 Mathematical Modeling.....	9
3.3 Property Table.....	10
3.4 Design Calculation.....	11
3.4 Design Output	15
3.5 Rating Calculation	15
3.6 Rating Output	19
3.7 Parametric Study.....	20
Chapter 4: Design Strategy	24
4.1 Design Factors	24
4.2 Design Models.....	25
4.3 Fabricated Project	27
Chapter 5: Fabrication Method.....	29
5.1 Process of Fabrication.....	29
5.2 Cost Analysis.....	30
5.3 Pressure Testing	31
Conclusion	33
Reference.....	33

Acknowledgement

We extend our heartfelt thanks to our Teachers, Priom Das (Lecturer, Department of Mechanical Engineering, BUET), Md. Aminul Islam (Lecturer, Department of Mechanical Engineering, BUET) and Tanvir Anam Tipu (Teaching Assistant, Department of Mechanical Engineering, BUET), for their invaluable guidance, unwavering support, and insightful feedback throughout the duration of this project. Their expertise and encouragement have been instrumental in shaping the direction of our project.

We are also deeply thankful to our institution, Bangladesh University of Engineering and Technology, for providing us with the necessary resources, facilities, and academic environment that facilitated our learning and project work. The constant encouragement and support from our faculty members and department staff have been immensely appreciated.

Abstract

The purpose of the project was to build a gasketed-plate heat exchanger that is highly efficient and compact heat transferring device between two fluids. In this project, we explored the design, fabrication, and performance evaluation of a plate heat exchanger for various industrial applications.

The design phase involved selecting suitable materials, determining the heat exchanger's dimensions, and optimizing the plate arrangement to achieve the desired thermal performance. We employ computational tools and simulation software to predict heat transfer characteristics and pressure drop within the exchanger.

To build the heat exchanger plates precisely, fabrication procedures were investigated with an emphasis on material compatibility, corrosion resistance, and ease of maintenance. To improve heat transfer efficiency, particular attention was paid to plate shape and surface treatments.

In conclusion, this heat transfer project improved our knowledge of heat transfer phenomena and offers feasible methods to increase heat transfer effectiveness in a variety of applications. The knowledge gained from this project helped to advance the objective of improving energy effectiveness and thermal performance in engineering systems.

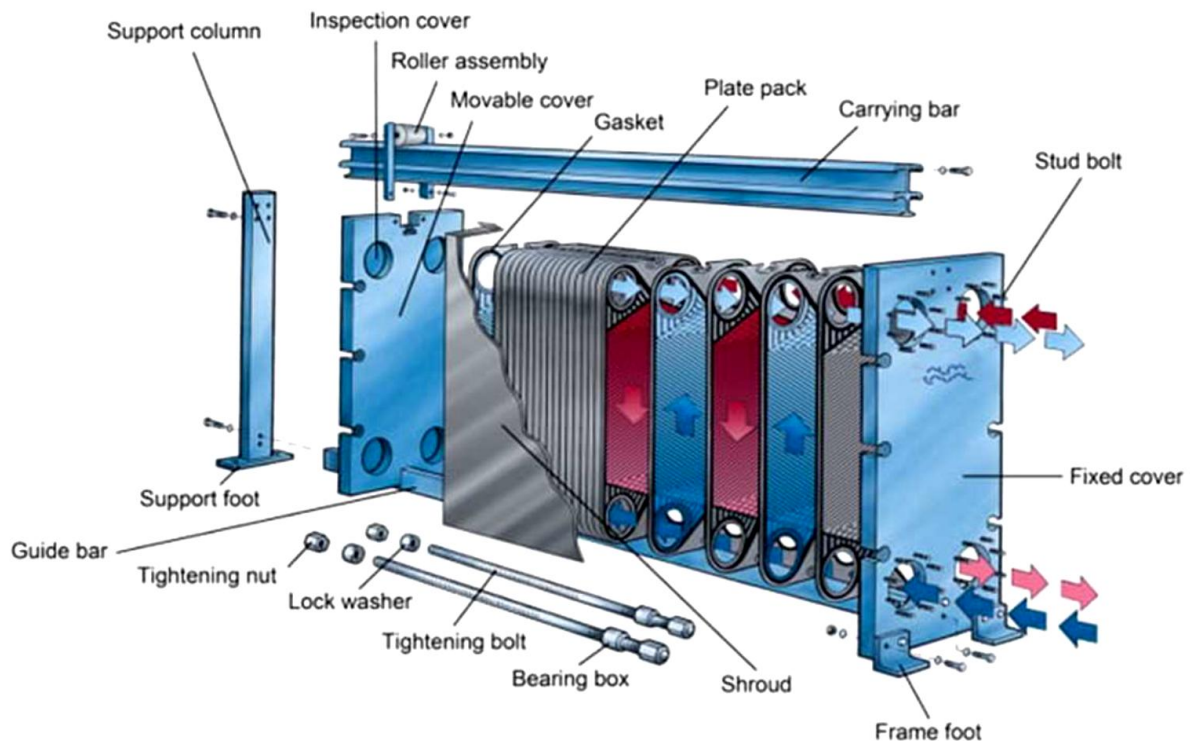
Chapter 1: Introduction

1.1 Introduction to Gasketed-Plate Heat Exchanger

A gasketed plate heat exchanger (GPHX) is a type of heat exchanger that uses a series of flat plates to transfer heat between two fluids. The plates are held together by gaskets, which create a seal between the plates and prevent the fluids from mixing.

Here is how a gasketed plate heat exchanger works:

1. The two fluids to be heat exchanged are pumped into the heat exchanger.
2. The fluids flow through the plates, which are arranged in a stack.



3. The heat is transferred from the hot fluid to the cold fluid through the plates.
4. The fluids exit the heat exchanger and are ready to be used.

The gaskets in a GPHX are an important part of the design. They create a seal between the plates and prevent the fluids from mixing. The gaskets are also designed to withstand the pressure and temperature of the fluids.

The versatility, energy efficiency, and space-saving attributes of GPHXs make them a compelling choice for a wide range of thermal management needs. By delving into the intricacies of their design and operation, we can harness their full potential and explore the many ways they contribute to enhanced efficiency and sustainability in various industrial processes and applications.

GPHXs are used in industrial, commercial, and residential heating and cooling systems. They are also used in food processing, chemical processing, and power generation.

1.2 Problem Statement

In a food processing plant 0.6 kg/s water at 50 °C is to be cooled using 0.6 kg/s water available at 20 °C. Targeted fluid should be reduced to 45 °C at least. Dimension of heat exchanger plate should be L=14 in, W=8 in.

Thickness of plate= 0.8mm. Our target is to determine the final temperature of hot water and number of plates needed for the heat exchanger. We assume fouling factor 0.00009 m²K/W.

1.3 Objectives

The objectives of this project are to-

- Improve the efficiency of heat transfer within the plate heat exchanger to maximize the exchange of thermal energy.
- Reduce energy consumption by optimizing the design and operation of the heat exchanger, leading to cost savings and environmental benefits.
- Enhance the reliability and durability of the heat exchanger to minimize downtime and maintenance requirements.
- Ensure that the materials and design of the heat exchanger are compatible with the specific fluids being processed to prevent corrosion or contamination.
- Identify opportunities to reduce the overall cost of the project, including installation, maintenance, and operation, while maintaining or improving performance.

Chapter 2: Overview of the Project

2.1 Specifications

- Hot water is cooled using cold water from 50 °C to 45 °C.
- We used water as both hot and cold fluid in this project.
- Counter flow design was used in this project. In this type, cold fluid passes in the opposite direction of hot fluid.
- Corrugated Aluminium plates were used in the GPHX.
- Rubber type gaskets were used in the heat exchanger.
- Leakage proof design.

2.2 Operating Principles of a Gasketed-Plate Heat Exchangers

A gasketed-plate heat exchanger is a type of heat exchanger that uses a series of plates to transfer heat between two fluids. The plates are made of a thin, corrugated material that provides a large surface area for heat transfer. The fluids flow through the plates in alternating channels, and the gaskets between the plates seal the channels and prevent the fluids from mixing. The hot fluid enters the heat exchanger at one end and flows through the channels in the plates. The cold fluid enters the heat exchanger at the other end and flows through the channels in the plates. The fluids flow in opposite directions, so that the hot fluid transfers heat to the cold fluid. The heat transfer is enhanced by the corrugated surface of the plates, which creates turbulence and increases the contact area between the fluids. The fluids exit the heat exchanger at a lower temperature than when they entered.

2.3 Advantages of Gasketed-plate Heat Exchanger

Gasketed plate heat exchangers are widely used in various industries for efficient heat transfer between two fluids. They offer numerous advantages, making them a popular choice for heating, cooling, and heat recovery applications. Here are 25 advantages of gasketed plate heat exchangers with descriptions:

1. **High Heat Transfer Efficiency:** Gasketed plate heat exchangers provide excellent heat transfer rates due to their large surface area and turbulent flow channels, resulting in efficient energy utilization.
2. **Compact Design:** They have a compact footprint compared to other heat exchanger types, making them suitable for space-constrained installations.
3. **Modular Construction:** Gasketed plates can be added or removed easily to adjust the heat exchanger's capacity, making them highly adaptable to changing process requirements.

4. **Temperature Control:** Gasketed plate heat exchangers allow precise temperature control by controlling flow rates and adjusting the number of plates.
5. **Easy Maintenance:** The removable gaskets and plates simplify maintenance, allowing for quick and cost-effective replacements.
6. **Reduced Fouling:** The turbulent flow in the plates minimizes fouling, reducing the need for frequent cleaning and maintenance.
7. **High Thermal Efficiency:** Their efficient design minimizes heat loss and maximizes heat recovery, resulting in energy savings.
8. **Corrosion Resistance:** Gasketed plates are available in various materials, including stainless steel and titanium, making them suitable for corrosive applications.
9. **Versatility:** They can handle a wide range of fluids, making them versatile for various industries, including food processing, HVAC, chemical, and pharmaceutical.
10. **Rapid Start-Up and Shutdown:** Gasketed plate heat exchangers can reach operating temperatures quickly and are easy to shut down when not in use, reducing energy consumption.
11. **Low Pressure Drop:** They have relatively low pressure drops, reducing the need for high-powered pumps and saving energy.
12. **Cost-Effective:** Gasketed plate heat exchangers are competitively priced, making them a cost-effective choice for many applications.
13. **Scalability:** They can be easily scaled up or down to accommodate changing production demands.
14. **High Flow Rate Capability:** Gasketed plate heat exchangers can handle high flow rates, making them suitable for demanding industrial processes.
15. **Enhanced Turbulence:** The corrugated design of plates creates turbulence, enhancing heat transfer efficiency.
16. **Reduced Maintenance Downtime:** Quick gasket and plate replacements minimize downtime during maintenance.
17. **Environmentally Friendly:** Their energy efficiency and ability to recover waste heat contribute to a reduced environmental footprint.
18. **Uniform Temperature Distribution:** Gasketed plates promote even distribution of temperature across the entire surface area.
19. **Reliable Performance:** Gasketed plate heat exchangers have a proven track record of reliability in various industries.
20. **Suitable for High-Temperature Applications:** They can handle high-temperature fluids, making them suitable for demanding industrial processes.

- 21. Long Service Life:** When properly maintained, gasketed plate heat exchangers can have a long service life, providing a good return on investment.
- 22. Customizable Design:** Plates can be configured to meet specific process requirements, such as counterflow or parallel flow.
- 23. Lower Fouling Risk:** Their design reduces the risk of fouling and clogging compared to other heat exchanger types.
- 24. Heat Recovery:** Gasketed plate heat exchangers are effective in recovering heat from waste streams, improving overall energy efficiency.
- 25. Easy Installation:** Their compact design and ease of integration into existing systems make them simple to install and retrofit into various applications.

These advantages collectively make gasketed plate heat exchangers a popular choice for many industries seeking efficient and cost-effective heat transfer solutions.

2.4 Limitations and Possible Solutions

Gasketed plate heat exchangers offer numerous advantages, but like any technology, they also have limitations. Here are some common limitations and possible solutions for gasketed plate heat exchangers:

- **Limited Temperature and Pressure Range:** Gasketed plate heat exchangers have temperature and pressure limitations depending on the materials used for the plates and gaskets. They may not be suitable for extreme high-temperature or high-pressure applications.
Solution: For high-temperature applications, consider using heat-resistant materials like stainless steel or alloys. For high-pressure applications, choose thicker plates and robust gaskets designed to withstand the pressure.
- **Risk of Gasket Leakage:** Over time, gaskets may deteriorate or develop leaks, leading to cross-contamination between the hot and cold fluids.
Solution: Regular inspection and maintenance are crucial. Replace gaskets periodically, and select gaskets made from materials compatible with the fluids and temperatures involved. Monitoring pressure and temperature differentials can help identify leaks early.
- **Fouling and Scaling:** Fouling, scaling, or sediment buildup on the plate surfaces can reduce heat transfer efficiency and increase maintenance requirements.
Solution: Implement proper water treatment and filtration systems to minimize fouling. Periodic cleaning, such as backflushing or chemical cleaning, can help maintain efficiency. Also, by increasing flow rate fouling can be reduced.

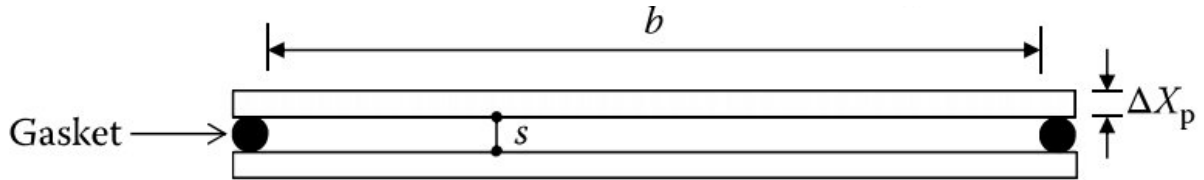
- Thermal Expansion and Stress:** Gasketed plate heat exchangers can experience thermal expansion and stress due to large temperature differentials, potentially leading to plate deformation.
Solution: Design the system to accommodate thermal expansion, use materials with low thermal expansion coefficients, and employ expansion joints if necessary.
- Compatibility Issues:** Compatibility between the materials of construction and the fluids being processed is essential to prevent corrosion and degradation.
Solution: Ensure proper material selection based on the fluid properties and operating conditions. Consider using corrosion-resistant coatings or linings if needed.
- Pressure Drop:** Gasketed plate heat exchangers can introduce pressure drops in the system due to the narrow flow channels.
Solution: Optimize the heat exchanger design for minimal pressure drop by selecting the appropriate plate configuration and ensuring a balanced flow distribution.

Chapter 3: Calculation & Optimization

3.1 Assumptions

- Steady operating condition has been achieved.
- Constant fluid properties
- Uniform fluid flow distribution
- No phase changes
- No heat transfer through gasket
- Constant flow rate

3.2 Mathematical Modeling



- Effective $NTU = F_{corr} NTU$, $F_{corr} = 1 - 0.0166 NTU$
- $NTU = \frac{U_0 A_0 N_s}{(\dot{m} C_p)_{min}}$, $A_0 = bl$
- $\frac{1}{U_0} = \frac{1}{h_h} + \frac{1}{h_c} + \frac{\Delta X_p}{k} + R_{di} + R_{do}$
- Flow velocity, $V = \frac{2 \dot{m}}{\rho s b (N_s + 1)}$; odd number of plates

$$V = \frac{2 \dot{m}}{\rho s b (N_s + 2)}$$
; even number of plates

- $D_h = \frac{4(\text{cross-sectional area})}{\text{wetted perimeter}} = \frac{4sb}{2s+2b} \cong \frac{4sb}{2b} = 2s$, $Re = \frac{VD_h}{\nu}$
- Laminar flow: $Nu = 1.86 Gz^{\frac{1}{3}} \left(\frac{\mu_b}{\mu_w} \right)^{0.14}$
- Turbulent flow: $Nu = 0.374 Re^{0.668} Pr^{\frac{1}{3}}$;
- Friction factor, $f = \begin{cases} \frac{280}{Re} : 1 < Re < 10 \\ \frac{100}{Re^{0.589}} : 10 < Re < 100 \\ \frac{12}{Re^{0.183}} : Re > 100 \end{cases}$
- Pressure drop, $\Delta P = \frac{fL}{D_h} \left(\frac{1}{2} \rho V^2 \right) + 1.3 \left(\frac{1}{2} \rho V_p^2 \right) \cong \frac{fL}{D_h} \left(\frac{1}{2} \rho V^2 \right)$

3.3 Property Table

WaterSat

T (°C)	P (kPa)	ρ (kg/m ³)	C _p (J/kg.K)	K (W/m.K)	$\mu \cdot 10^5$ (kg/m.s)	Pr
5	0.8721	999.9	4205	0.571	1.519	11.2
10	1.2276	999.7	4194	0.58	1.307	9.45
15	1.7051	999.1	4185	0.589	1.138	8.09
20	2.339	998	4182	0.598	1.002	7.01
25	3.169	997	4180	0.607	0.891	6.14
30	4.246	996	4178	0.615	0.798	5.42
35	5.628	994	4178	0.623	0.72	4.83
40	7.384	992.1	4179	0.631	0.653	4.32
45	9.593	990.1	4180	0.637	0.596	3.91
50	12.35	988.1	4181	0.644	0.547	3.55
55	15.76	985.2	4183	0.649	0.504	3.25
60	19.94	983.3	4185	0.654	0.467	2.99
65	25.03	980.4	4187	0.659	0.433	2.75
70	31.19	977.5	4190	0.663	0.404	2.55
75	38.58	974.7	4193	0.667	0.378	2.38
80	47.39	971.8	4197	0.67	0.355	2.22
85	57.83	968.1	4201	0.673	0.333	2.08
90	70.14	965.3	4206	0.675	0.315	1.96
95	84.55	961.5	4212	0.677	0.297	1.85
100	101.33	957.9	4217	0.679	0.282	1.75
110	143.27	950.6	4229	0.682	0.255	1.58
120	198.53	943.4	4244	0.683	0.232	1.44
130	270.1	934.6	4263	0.684	0.231	1.33
140	361.3	921.7	4286	0.683	0.197	1.24
150	475.8	916.6	4311	0.682	0.183	1.16

3.4 Design Calculation

```
% Input
tci=20;
thi=50;
tho=45.55;
mh=0.6;
mc=0.6;
% Geometric Parameters and Properties
s=0.0035;
A=0.02306;
Ac=349.96*10^(-6);
Dh=6.756*10^(-3);
Dp=0.0254;
Rf=0.00009;
t=0.0008;
km=237;
% Thermophysical Properties of Hot Fluid
tavgh=(thi+tho)/2;
for i=1:1:24
    if tavgh==WaterSat{i,1}
        dh=WaterSat{i,3};
        cph=WaterSat{i,4};
        kh=WaterSat{i,5};
        miuh=WaterSat{i,6}*(10^-3);
        prh=WaterSat{i,7};
        break
    elseif tavgh>WaterSat{i,1} && tavgh<WaterSat{i+1,1}
        dh1=WaterSat{i,3};
        cph1=WaterSat{i,4};
        kh1=WaterSat{i,5};
        miuh1=WaterSat{i,6}*(10^-3);
        prh1=WaterSat{i,7};
        dh2=WaterSat{i+1,3};
        cph2=WaterSat{i+1,4};
        kh2=WaterSat{i+1,5};
        miuh2=WaterSat{i+1,6}*(10^-3);
        prh2=WaterSat{i+1,7};
        th1=WaterSat{i,1};
        th2=WaterSat{i+1,1};
        dh=((dh1-dh2)*(tavgh-th2))/(th1-th2)+dh2;
        cph=((cph1-cph2)*(tavgh-th2))/(th1-th2)+cph2;
        kh=((kh1-kh2)*(tavgh-th2))/(th1-th2)+kh2;
        miuh=((miuh1-miuh2)*(tavgh-th2))/(th1-th2)+miuh2;
        prh=((prh1-prh2)*(tavgh-th2))/(th1-th2)+prh2;
        break
    end
end
% Thermophysical Properties of Cold Fluid
q=mh*cph*(thi-tho);
tcoas=25;
tavgc=(tci+tcoas)/2;
for i=1:1:24
    if tavgc==WaterSat{i,1}
        cpc=WaterSat{i,4};
        break
    elseif tavgc>WaterSat{i,1} && tavgc<WaterSat{i+1,1}
        cpc1=WaterSat{i,4};
```

```

        cpc2=WaterSat{i+1,4};
        tc1=WaterSat{i,1};
        tc2=WaterSat{i+1,1};
        cpc=((cpc1-cpc2)*(tavgc-tc2))/(tc1-tc2)+cpc2;
        break
    end
end
q0=mc*cpc*(tcoas-tci);
if q0==q
    tco=tcoas;
else
    itr1=0;
    while 1
        tcoas=q/(mc*cpc)+tci;
        tavgc=(tci+tcoas)/2;
        for i=1:1:24
if tavgc==WaterSat{i,1}
            cpc=WaterSat{i,4};
            break
elseif tavgc>WaterSat{i,1} && tavgc<WaterSat{i+1,1}
                cpc1=WaterSat{i,4};
                cpc2=WaterSat{i+1,4};
                tc1=WaterSat{i,1};
                tc2=WaterSat{i+1,1};
                cpc=((cpc1-cpc2)*(tavgc-tc2))/(tc1-tc2)+cpc2;
                break
            end
                end
                q0=mc*cpc*(tcoas-tci);
                itr1=itr1+1;
                if abs(q0-q)>0 && abs(q0-q)<0.5
                    tco=tcoas;
                    break
                end
            end
        end
        tavgc=(tci+tco)/2;
for i=1:1:24
    if tavgc==WaterSat{i,1}
        dc=WaterSat{i,3};
        cpc=WaterSat{i,4};
        kc=WaterSat{i,5};
        miuc=WaterSat{i,6}*(10^-3);
        prc=WaterSat{i,7};
        break
    elseif tavgc>WaterSat{i,1} && tavgc<WaterSat{i+1,1}
        dc1=WaterSat{i,3};
        cpc1=WaterSat{i,4};
        kc1=WaterSat{i,5};
        miuc1=WaterSat{i,6}*(10^-3);
        prc1=WaterSat{i,7};
        dc2=WaterSat{i+1,3};
        cpc2=WaterSat{i+1,4};
        kc2=WaterSat{i+1,5};
        miuc2=WaterSat{i+1,6}*(10^-3);
        prc2=WaterSat{i+1,7};
        tc1=WaterSat{i,1};
        tc2=WaterSat{i+1,1};
        dc=((dc1-dc2)*(tavgc-tc2))/(tc1-tc2)+dc2;
    end
end

```

```

        cpc=((cpc1-cpc2)*(tavgc-tc2))/(tc1-tc2)+cpc2;
        kc=((kc1-kc2)*(tavgc-tc2))/(tc1-tc2)+kc2;
        miuc=((miuc1-miuc2)*(tavgc-tc2))/(tc1-tc2)+miuc2;
        prc=((prc1-prc2)*(tavgc-tc2))/(tc1-tc2)+prc2;
        break
    end
end
% Calculation of Required Number of Plates and Thermal Performance
tmax=thi-tci;
cc=mc*cpc;
ch=mh*cph;
if cc>ch
    cmax=cc;
    cmin=ch;
else
    cmin=cc;
    cmax=cc;
end
qmax=cmin*tmax;
e=(q/qmax)*100;
c=cmin/cmax;
syms ntu0 c0 e0
ntu1=0.25;
e=e/100;
er=0.001;
itr2=0;
if c==1
    f=ntu0/(1+ntu0)-e0;
else
    f=(1-exp(-ntu0*(1-c0)))/(1-c0*exp(-ntu0*(1-c0)))-e0;
end
g=diff(f,ntu0);
df_f=double(subs(g,{ntu0,c0,e0},{ntu1,c,e}));
if df_f==0
    disp('NTU can not be determined');
else
    while 1
        itr2=itr2+1;
        p=double(subs(f,{ntu0,c0,e0},{ntu1,c,e}));
        if p==0
            break
        end
        df_f=double(subs(g,{ntu0,c0,e0},{ntu1,c,e}));
        ntu2=ntu1-p/df_f;
        if abs((ntu2-ntu1)/ntu2)<=er
            ntu1=ntu2;
            break
        end
        ntu1=ntu2;
    end
end
fcorr=1-0.0166*ntu1;
ntu1=fcorr*ntu1;
UN=(ntu1*cmin)/A;
N=12;
itr3=0;
while 1
    itr3=itr3+1;
    U=UN/N;

```

```

N=ceil(N);
if rem(N,2)==0
    vh=(2*mh)/(dh*Ac*(N+2));
    vc=(2*mc)/(dc*Ac*(N+2));
else
    vh=(2*mh)/(dh*Ac*(N+1));
    vc=(2*mc)/(dc*Ac*(N+1));
end
Reh=(dh*Dh*vh)/miuh;
Rec=(dc*Dh*vc)/miuc;
if Reh<100
    Gzh=(Reh*Prh)/(1/Dh);
    Nuh=1.86*(Gzh)^(1/3);
else
    Nuh=0.374*(Reh^0.668)*(prh^(1/3));
end
if Rec<100
    Gzc=(Rec*Prc)/(1/Dh);
    Nuc=1.86*(Gzc)^(1/3);
else
    Nuc=0.374*(Rec^0.668)*(prc^(1/3));
end
hh=(Nuh*kh)/Dh;
hc=(Nuc*kc)/Dh;
U0=(1/hc+1/hh+2*Rf)^-1;
if abs((U0-U)/U)<=0.01
    Ns=N;
    U1=UN/Ns;
    break
end
N=UN/U0;
end
% Output
disp('All values are in SI unit');
fprintf('ph=%.4f\n',dh);
fprintf('cph=%.4f\n',cph);
fprintf('kh=%.4f\n',kh);
fprintf('μh=%.4f\n',miuh);
fprintf('Prh=%.4f\n',prh);
fprintf('pc=%.4f\n',dc);
fprintf('cpc=%.4f\n',cpc);
fprintf('kc=%.4f\n',kc);
fprintf('μc=%.4f\n',miuc);
fprintf('Prc=%.4f\n',prc);
fprintf('hh=%.4f\n',hh);
fprintf('hc=%.4f\n',hc);
fprintf('Uo=%.4f\n',U1);
fprintf('Q=%.4f\n',q);
fprintf('Tco=%.4f\n',tco);
fprintf('Required number of plates=%.4f\n',Ns);

```

3.4 Design Output

```
Command Window
>> Design_PFH
All values are in SI unit
ph=988.9900
cph=4180.5550
kh=0.6409
μh=0.0006
Prh=3.7102
ρc=997.5550
cpc=4181.1101
kc=0.6020
μc=0.0010
Prc=6.6229
hh=16440.0436
hc=13274.1355
Uo=3148.3376
Q=11162.0819
Tco=24.4495
Required number of plates=6.0000
fx >>
```

3.5 Rating Calculation

```
% Input
tci=20;
thi=50;
mh=0.6;
mc=0.6;
% Geometric Parameters and Properties
s=0.0035;
A=0.02306;
Ac=349.96*10^(-6);
Dh=6.756*10^(-3);
Dp=0.0254;
N=6;
Rf=0.00009;
t=0.0008;
km=237;
thoas=38;
tcoas=23;
while 1
    % Thermophysical Properties of Hot Fluid
    tavgh=(thi+thoas)/2;
for i=1:1:24
    if tavgh==WaterSat{i,1}
        dh=WaterSat{i,3};
        cph=WaterSat{i,4};
        kh=WaterSat{i,5};
        miuh=WaterSat{i,6}*(10^-3);
        prh=WaterSat{i,7};
        break
    elseif tavgh>WaterSat{i,1} && tavgh<WaterSat{i+1,1}
        dh1=WaterSat{i,3};
        cph1=WaterSat{i,4};
        kh1=WaterSat{i,5};
        miuh1=WaterSat{i,6}*(10^-3);
        prh1=WaterSat{i,7};
```

```

        dh2=WaterSat{i+1,3};
        cph2=WaterSat{i+1,4};
        kh2=WaterSat{i+1,5};
        miuh2=WaterSat{i+1,6}*(10^-3);
        prh2=WaterSat{i+1,7};
        th1=WaterSat{i,1};
        th2=WaterSat{i+1,1};
        dh=((dh1-dh2)*(tavgh-th2))/(th1-th2)+dh2;
        cph=((cph1-cph2)*(tavgh-th2))/(th1-th2)+cph2;
        kh=((kh1-kh2)*(tavgh-th2))/(th1-th2)+kh2;
        miuh=((miuh1-miuh2)*(tavgh-th2))/(th1-th2)+miuh2;
        prh=((prh1-prh2)*(tavgh-th2))/(th1-th2)+prh2;
        break
    end
end
if rem(N,2)==0
    vh=(2*mh)/(dh*Ac*(N+2));
else
    vh=(2*mh)/(dh*Ac*(N+1));
end
Reh=(dh*Dh*vh)/miuh;
if Reh<100
    Gzh=(Reh*Prh)/(1/Dh);
    Nuh=1.86*(Gzh)^(1/3);
else
    Nuh=0.374*(Reh^0.668)*(prh^(1/3));
end
hh=(Nuh*kh)/Dh;
% Thermophysical Properties of Cold Fluid
tavgc=(tci+tcoas)/2;
for i=1:1:24
    if tavgc==WaterSat{i,1}
        dc=WaterSat{i,3};
        cpc=WaterSat{i,4};
        kc=WaterSat{i,5};
        miuc=WaterSat{i,6}*(10^-3);
        prc=WaterSat{i,7};
        break
    elseif tavgc>WaterSat{i,1} && tavgc<WaterSat{i+1,1}
        dc1=WaterSat{i,3};
        cpc1=WaterSat{i,4};
        kc1=WaterSat{i,5};
        miuc1=WaterSat{i,6}*(10^-3);
        prc1=WaterSat{i,7};
        dc2=WaterSat{i+1,3};
        cpc2=WaterSat{i+1,4};
        kc2=WaterSat{i+1,5};
        miuc2=WaterSat{i+1,6}*(10^-3);
        prc2=WaterSat{i+1,7};
        tc1=WaterSat{i,1};
        tc2=WaterSat{i+1,1};
        dc=((dc1-dc2)*(tavgc-tc2))/(tc1-tc2)+dc2;
        cpc=((cpc1-cpc2)*(tavgc-tc2))/(tc1-tc2)+cpc2;
        kc=((kc1-kc2)*(tavgc-tc2))/(tc1-tc2)+kc2;
        miuc=((miuc1-miuc2)*(tavgc-tc2))/(tc1-tc2)+miuc2;
        prc=((prc1-prc2)*(tavgc-tc2))/(tc1-tc2)+prc2;
        break
    end
end
end

```

```

if rem(N,2)==0
    vc=(2*mh)/(dc*Ac*(N+2));
else
    vc=(2*mh)/(dc*Ac*(N+1));
end
Rec=(dc*Dh*vc)/miuc;
if Rec<100
    Gzc=(Rec*Prh)/(1/Dh);
    Nuc=1.86*(Gzc)^(1/3);
else
    Nuc=0.374*(Rec^0.668)*(prc^(1/3));
end
hc=(Nuc*kc)/Dh;
%Thermal Performance
U0=(1/hc+1/hh+2*Rf)^-1;
tmax=thi-tci;
cc=mc*cpc;
ch=mh*cph;
if cc>ch
    cmax=cc;
    cmin=ch;
else
    cmin=cc;
    cmax=cc;
end
qmax=cmin*tmax;
c=cmin/cmax;
entu=(U0*A*N)/cmin;
syms ntu0 entu0
ntu1=0.4;
er=0.001;
f=(ntu0-0.0166*(ntu0^2))-entu0;
g=diff(f,ntu0);
df_f=double(subs(g,{ntu0,entu0},{ntu1,entu}));
if df_f==0
    disp('NTU can not be determined');
else
    while 1
        p=double(subs(f,{ntu0,entu0},{ntu1,entu}));
        if p==0
            break
        end
        df_f=double(subs(g,{ntu0,entu0},{ntu1,entu}));
        ntu2=ntu1-p/df_f;
        if abs((ntu2-ntu1)/ntu2)<=er
            ntu1=ntu2;
            break
        end
        ntu1=ntu2;
    end
end
ntu=ntu1;
if c==1
    e=ntu/(1+ntu);
else
    e=(1-exp(-ntu*(1-c)))/(1-c*exp(-ntu*(1-c)));
end
q=e*qmax;
tco=tci+q/cc;

```

```

tho=thi-q/ch;
if abs((tho-thoas)/thoas)<0.0001 && abs((tco-tcoas)/tcoas)<0.0001
    break
end
thoas=tho;
tcoas=tco;
end
if Reh>1 && Reh<10
    f=280/Reh;
elseif Reh>10 && Reh<100
    f=100/(Reh^0.589);
else
    f=12/(Reh^0.183);
end
% Pressure Drop
vph=mh/(dh*((pi*Dp*Dp)/4));
DelPh=(f*l/Dh)*(0.5*dh*vh*vh)+1.3*(0.5*dh*vph*vph);
if Rec>1 && Rec<10
    f=280/Rec;
elseif Rec>10 && Rec<100
    f=100/(Rec^0.589);
else
    f=12/(Rec^0.183);
end
vpc=mc/(dc*((pi*Dp*Dp)/4));
DelPc=(f*l/Dh)*(0.5*dc*vc*vc)+1.3*(0.5*dc*vpc*vpc);
% Output
disp('All values are in SI unit');
fprintf('ph=%.4f\n',dh);
fprintf('cph=%.4f\n',cph);
fprintf('kh=%.4f\n',kh);
fprintf('μh=%.4f\n',miuh);
fprintf('Prh=%.4f\n',prh);
fprintf('pc=%.4f\n',dc);
fprintf('cpc=%.4f\n',cpc);
fprintf('kc=%.4f\n',kc);
fprintf('μc=%.4f\n',miuc);
fprintf('Prc=%.4f\n',prc);
if Reh<100
    disp('Hot Water flow is laminar');
else
    disp('Hot Water flow is turbulent');
end

if Rec<100
    disp('Cold Water flow is laminar');
else
    disp('Cold Water flow is turbulent');
end
end
fprintf('hh=%.4f\n',hh);
fprintf('hc=%.4f\n',hc);
fprintf('Uo=%.4f\n',U0);
fprintf('Effectiveness=%.4f\n',e);
fprintf('NTU=%.4f\n',ntu);
fprintf('Q=%.4f\n',q);
fprintf('Tho=%.4f\n',tho);
fprintf('Tco=%.4f\n',tco);
fprintf('Pressure Drop of Hot Fluid=%.4f\n',DelPh);
fprintf('Pressure Drop of Cold Fluid=%.4f\n',DelPc);

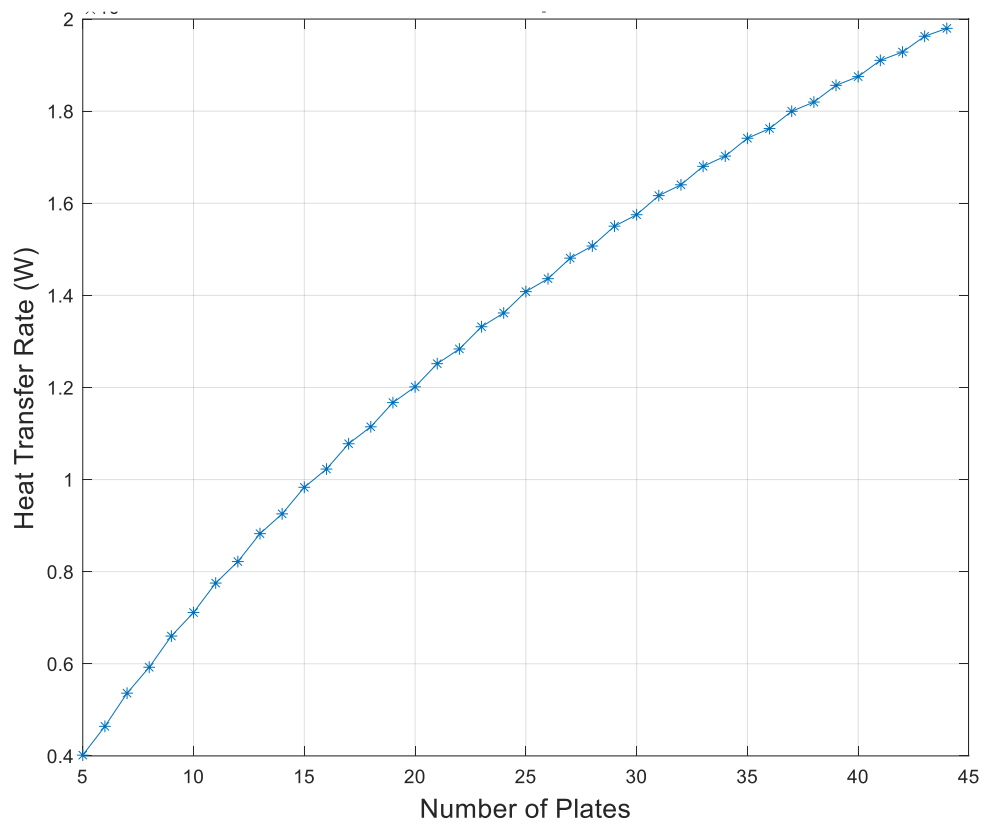
```

3.6 Rating Output

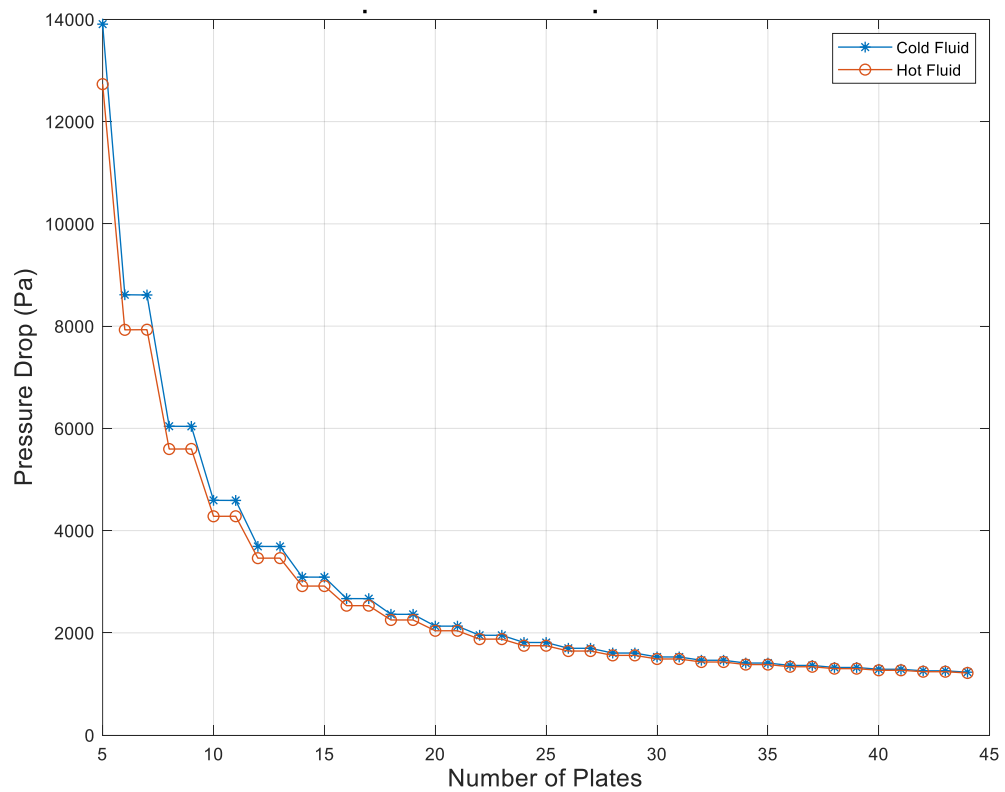
```
Command Window
>> Rating_PFHX
All values are in SI unit
ph=988.9935
cph=4180.5532
kh=0.6409
μh=0.0006
Prh=3.7108
ρc=997.5533
cpc=4181.1066
kc=0.6020
μc=0.0010
Prc=6.6214
Hot Water flow is turbulent
Cold Water flow is turbulent
hh=16438.9987
hc=13275.2723
Uo=3162.9657
Effectiveness=0.1489
NTU=0.1750
Q=11206.3493
Tho=45.5323
Tco=24.4671
Pressure Drop of Hot Fluid=7952.3368
Pressure Drop of Cold Fluid=8573.4233
fx >>
```

3.7 Parametric Study

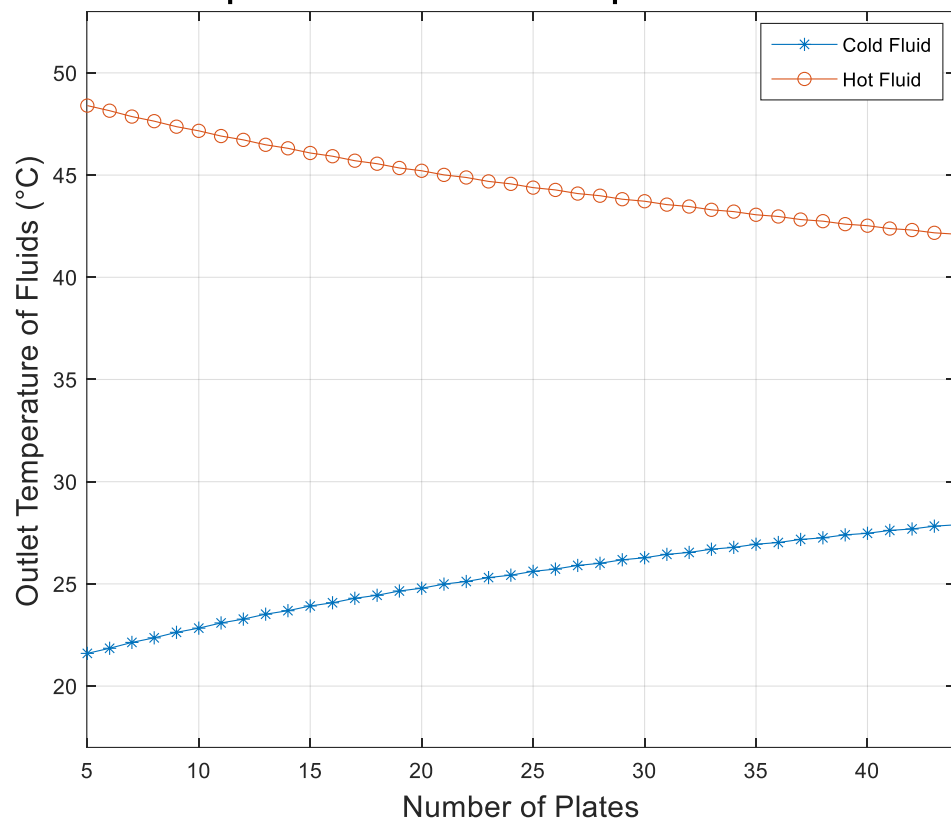
Variation of Heat Transfer Rate with Number of Plates



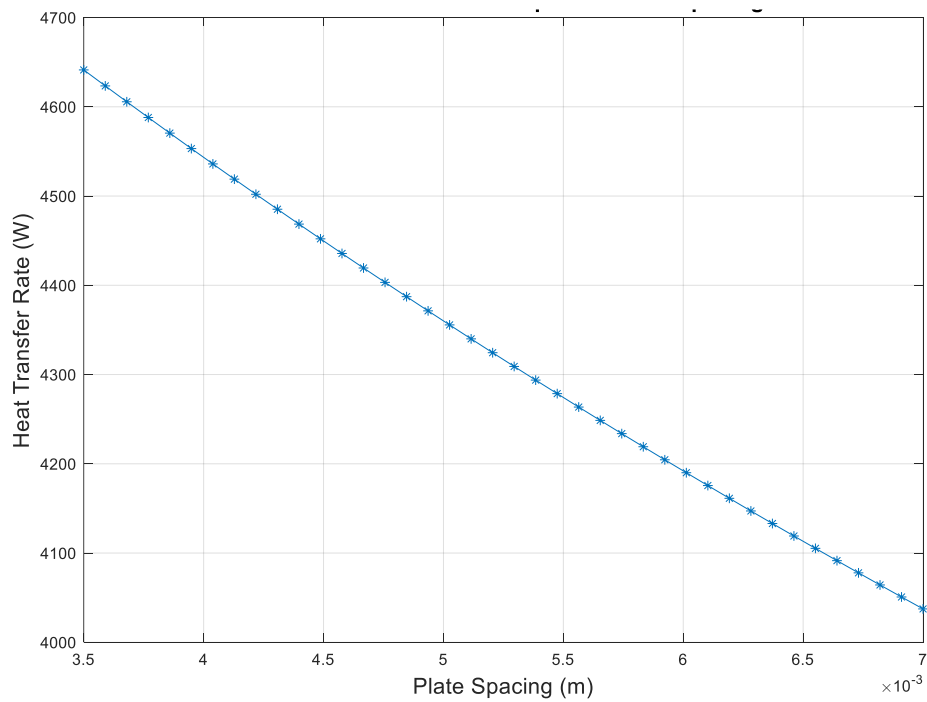
Variation of Pressure Drop of Fluids with Numbers of Plates



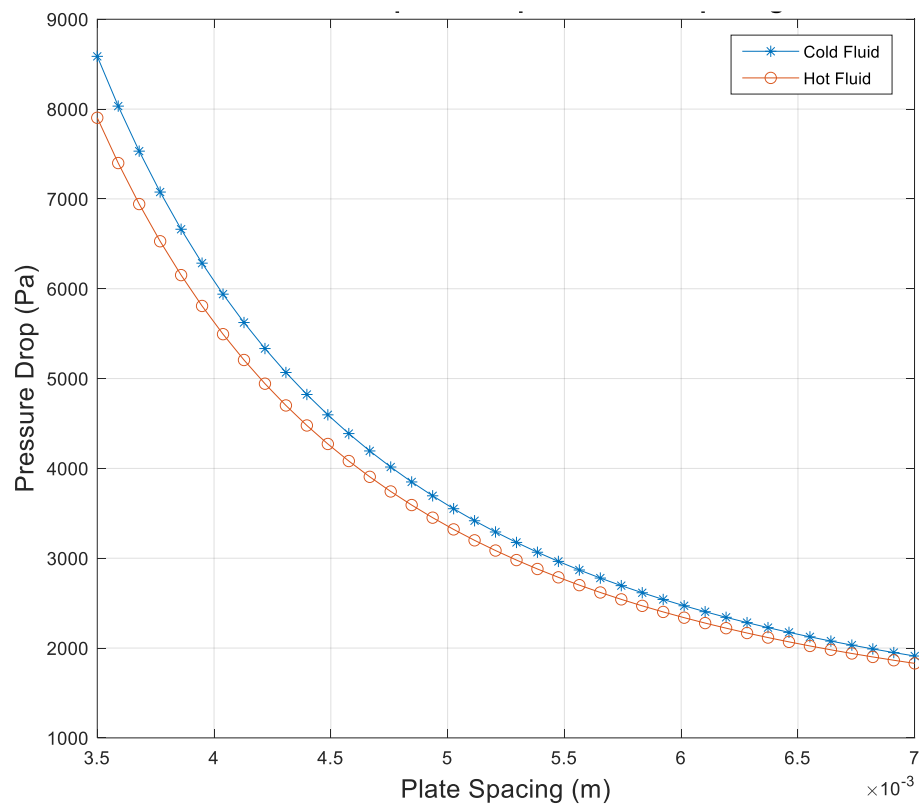
Variation of Outlet Temperature of Fluids with Number of Plates



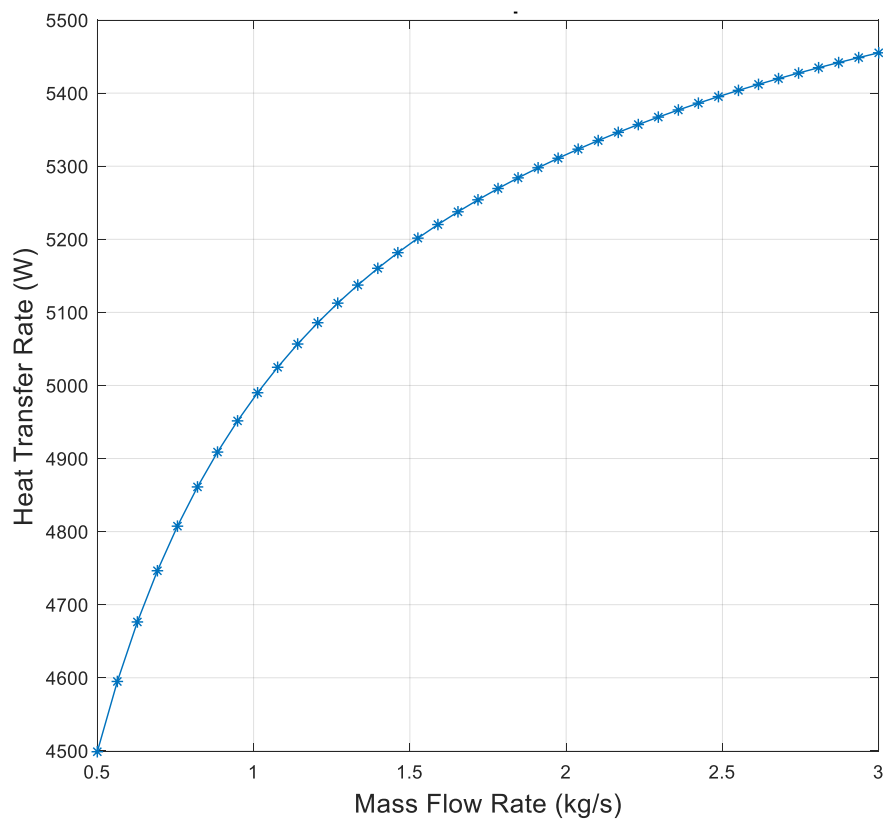
Variation of Heat Transfer Rate with Plate Spacing



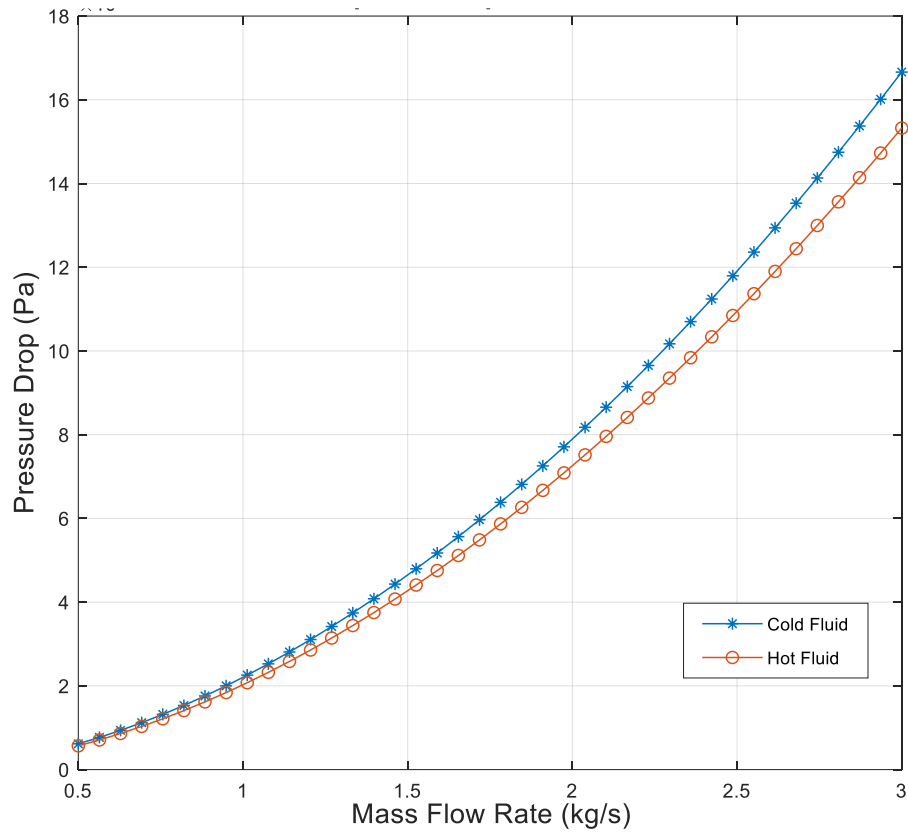
Variation of Pressure Drop with Plate Spacing



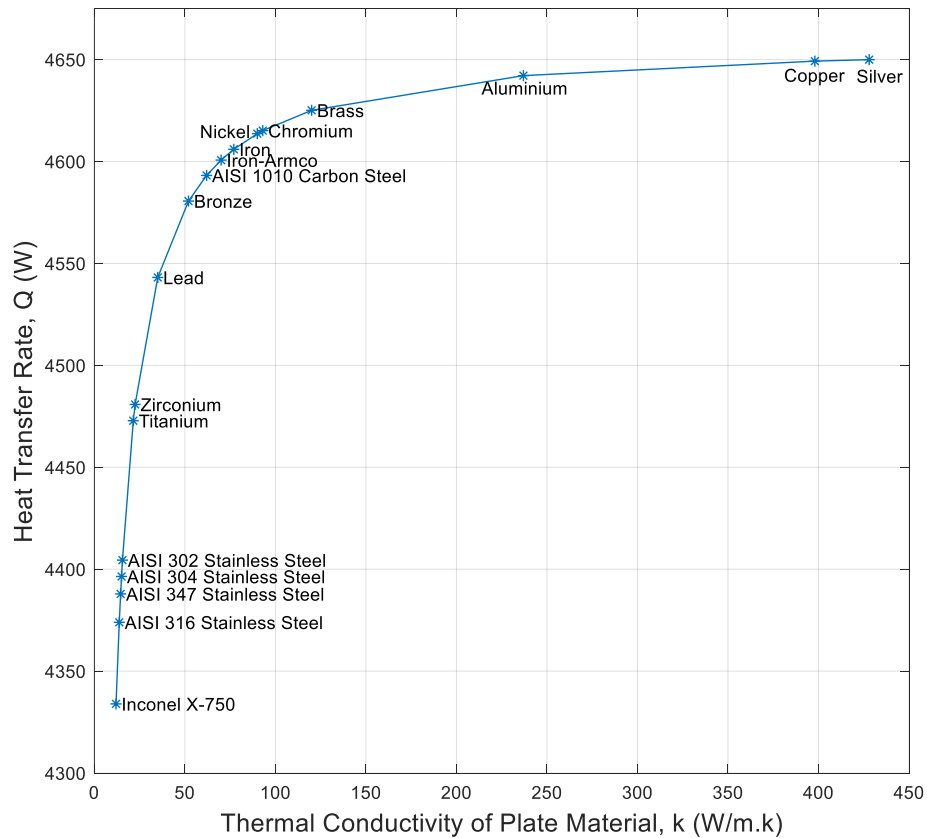
Variation of Heat Transfer Rate with Mass Flow Rate



Variation of Pressure Drop with Mass Flow Rate



Variation of Heat Transfer Rate with Thermal Conductivity of Plate Material

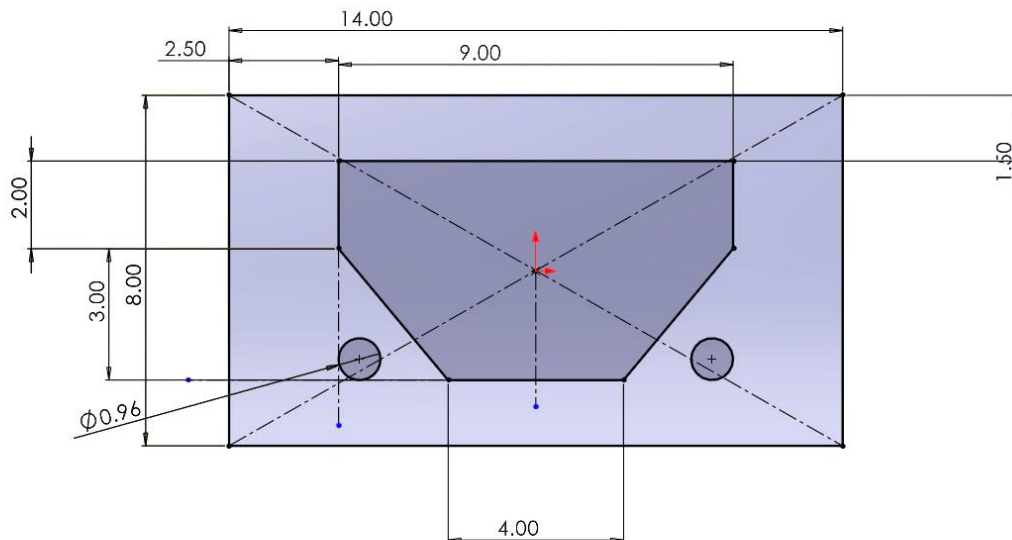


Chapter 4: Design Strategy

4.1 Design Factors

Dimensions:

- Plate thickness = 1 mm
- Gasket thickness = 5 mm
- No. of plates = 6



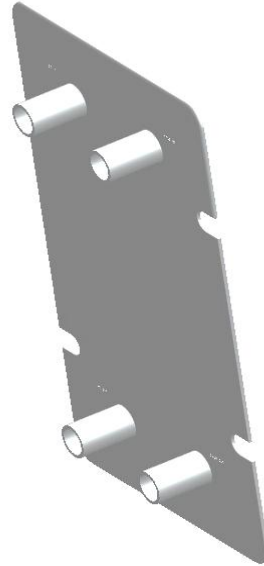
Material used: Aluminum plates.

4.2 Design Models

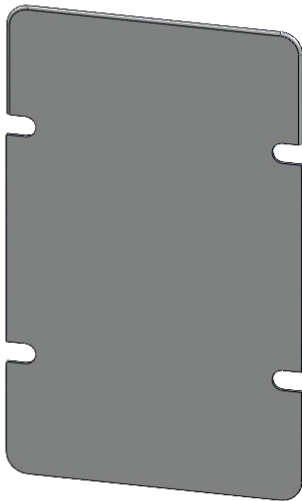
Software used: SolidWorks.



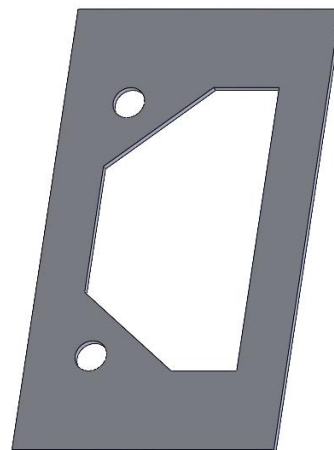
Plate



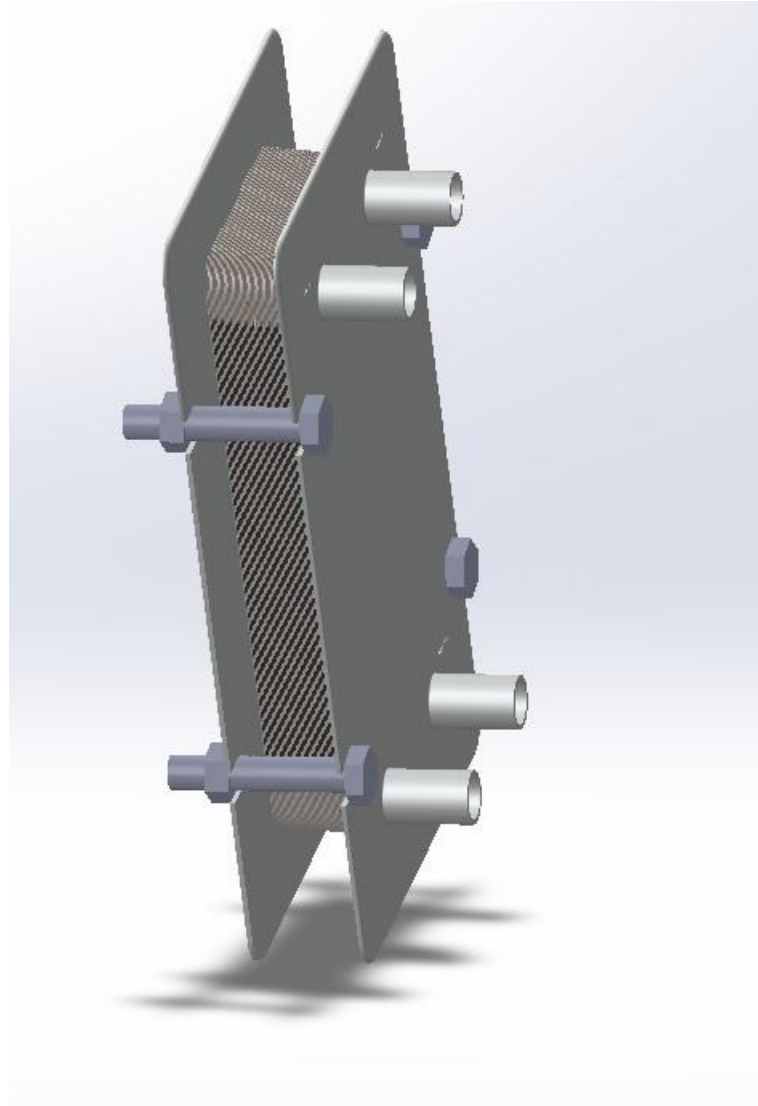
Front Cover



back cover



Gasket



Isometric View

4.3 Fabricated Project



Corrugated Plate



Gasket



Screw&Bolt



Pipe



Assembled GPHX

Chapter 5: Fabrication Method

5.1 Process of Fabrication

For the fabrication of a gasketed plate heat exchanger we followed several steps to create the plates, assemble the heat exchanger, and ensure its quality and functionality. Here is an overview of the process:

Design and Specification: At first, we determined the heat exchanger's design parameters, including heat transfer requirements, flow rates, materials, and dimensions. Then, we created detailed engineering drawings and specifications for the heat exchanger.

Material Selection: We chose appropriate materials for the plates, gaskets, frame, and other components based on the process requirements and compatibility with the fluids being handled.

Plate Manufacturing: For manufacture the heat transfer plates we selected Aluminum metal sheet. The plates were corrugated to enhance turbulence and heat transfer efficiency. Precision manufacturing processes were used to ensure the plates' uniformity and quality.

Frame and Connections: Then we fabricated the frame and connecting components that held the plates together.

The frame provides support and maintains the correct alignment of the plates. Besides, we installed inlet and outlet connections, as well as ports for the two fluid streams.

Drilling: Drilling was carried out to create ports on the plates and front side of the frame to pass liquid from the inlets to the outlets.

Grinding: We also performed grinding to achieve a smooth surface and to remove dirt and scaling on the plates and frames.



Welding: We selected the appropriate welding method for the materials involved. Depending on the type of metal used for the pipes and the heat exchanger frame we perform metal Arc welding.

Gasket Manufacturing: We produced gaskets which were made of rubber materials that are suitable for sealing and withstanding the process conditions. Gaskets were planes to prevent fluid leakage.

Assembly: We assembled the heat exchanger by stacking the plates and gaskets together in an alternating pattern.

Here, we ensured the plates and gaskets were aligned correctly, and the frame was securely tightened to compress the plates and gaskets, creating a tight seal.

Quality Control and Testing: Then we performed quality control checks to verify that the heat exchanger meets design specifications and tolerances.

We often conducted pressure and leak tests to confirm the integrity of the seals and connections.

Final Inspection: At last, we conducted a final inspection of the assembled heat exchanger to confirm that all components are in place and functioning correctly.

We also verified that the heat exchanger meets safety and regulatory standards.

5.2 Cost Analysis

Description	Unit cost	Total Cost (BDT)
Mild steel sheet	110(per kg)	360
Al sheet	170 (per ft)	1000
Metal sheet cutting	-	250
Rubber sheet	25(per ft)	150
Joint pipe & fittings	100 (per pcs)	550
Drilling and grinding	-	700
Fasteners	40 (per pcs)	240
Glue gun	-	250
Transport	-	660
Total		4160

5.3 Pressure Testing

Pressure test was performed in Hydraulics Lab. Due to limitation only pressure test was done. To test the pressure limit, the outlets are closed using death cap and filled with water. And then pressure applied in inlet using hand pump. It started leaking after 10 psi. Leakage found in welding and gaskets.



Figure: Setup for the Pressure Test

Date: 10/09/2023

Project: Heat exchanger (Gasketed plate HX).

Group No: B-03

Group member: Asman Mia (1910117)
Sabit Hasnat Mahee (1910091)
Barakatullah Fahad (1910115)
Shahriar Parves (1910114).

pressure test:

The Heat Exchanger sustained at pressure of 10 psi. But when pressure was further increased, it failed and leakage happened through the gasket.



Md. Abul Kalam Azad
Senior Lab Instructor
Dept. of Mechanical Engineering
BUET, Dhaka-1000

Conclusion

We tried to design a compact Heat exchanger, so we chose Gasketed-Plate heat exchanger which has high heat transfer rate but needs smaller area for installation. Besides it is a cost-effective heat exchanger.

The heat exchanger shows high performance of heat transfer rate, lower pressure drop.

We faced some difficulties designing. Our plate was corrugated on one side only, as a result heat transfer area reduce as well as low temperature drop.

If we used CNC then corrugated plate might be done on both sides, but due to lack of availability of CNC in market, we used readymade corrugated surface one side only.

Except for some limitations gasketed plate heat exchangers (GPHXs) are versatile and efficient heat transfer devices used in a wide range of industries. They offer benefits such as high thermal efficiency, compact design, and ease of maintenance. Future enhancements in GPHX technology are likely to focus on materials, smart monitoring, energy efficiency, sustainability, modular design, compactness, heat recovery, computational tools, water treatment, and cost reduction.

Reference

1. Cengel, Y.A., & Ghajar, A.J. (2014). Heat and Mass Transfer: Fundamentals and Applications (5th ed.). McGraw-Hill Professional.
2. S. Kakaç, H. Liu, A. Pramuanjaroenkij, Heat Exchangers: Selection, Rating, and Thermal Design, Third Edition, CRC Press, 2012.